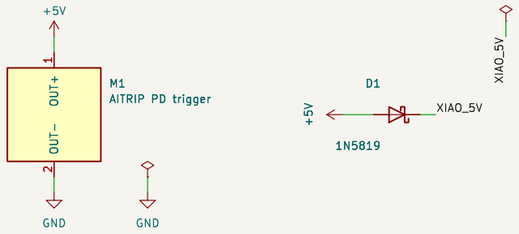
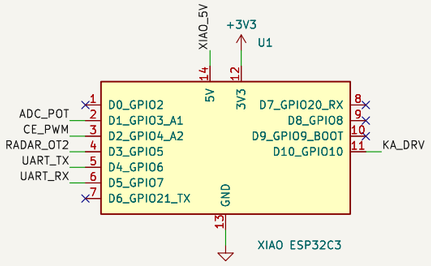
1 POWER ENTRY



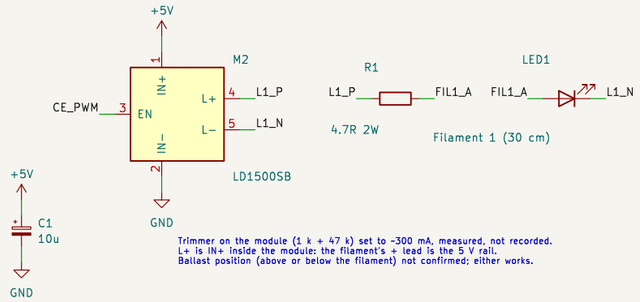
Bank USB-C -> PD trigger, set to 5 V and button-locked.
Verified to keep the BODQPD24ZD bank awake on its own.
No fuse anywhere (see design review P1).
1N5819 feeds the XIAO's 5V pin and blocks back-feed
when the XIAO's own USB-C is plugged in for flashing.

2 MCU: XIAO ESP32C3



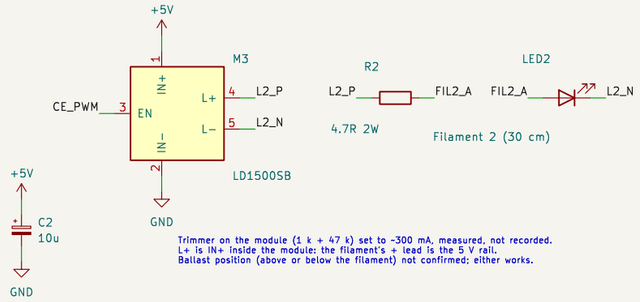
Firmware: src/main.c (a631148). D2 = 1 kHz PWM, 13-bit, XTAL clock.
ADC_RAW_AT_MAX = 2950 measured on this unit.

3 FILAMENT 1: LD1500SB (CN5711 LINEAR SINK)

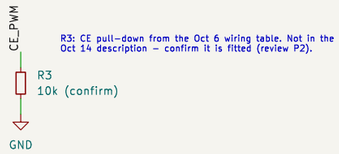


Trimmer on the module (1 k + 47 k) set to ~300 mA, measured, not recorded.
L+ is IN+ inside the module: the filament's + lead is the 5 V rail.
Ballast position (above or below the filament) not confirmed; either works.

4 FILAMENT 2: LD1500SB (CN5711 LINEAR SINK)

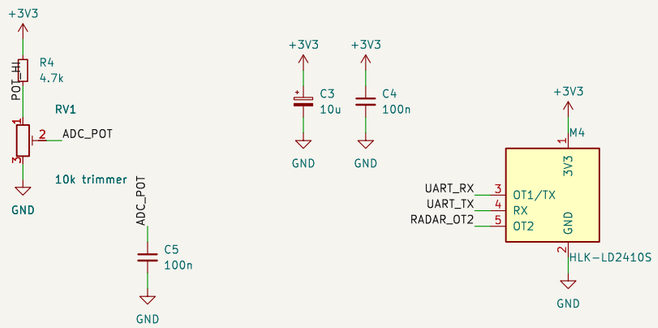


Trimmer on the module (1 k + 47 k) set to ~300 mA, measured, not recorded.
L+ is IN+ inside the module: the filament's + lead is the 5 V rail.
Ballast position (above or below the filament) not confirmed; either works.



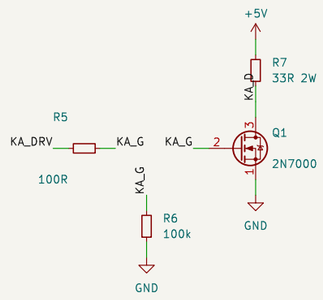
R3: CE pull-down from the Oct 6 wiring table. Not in the
Oct 14 description — confirm it is fitted (review P2).

5 BRIGHTNESS TRIMMER AND RADAR



Wiper tops out at 2.25 V (4.7 k above the 10 k trimmer).

6 BANK KEEP-ALIVE



152 mA nominal (~115 mA at a 3.3 V gate), 200 ms every 10 s
while the lamp is dark. 2N7000 flat face: S G D.

Superseded for production by Rev G1 (fgl_g1)
2 x LD1500SB (CN5711 linear) modules, current set by on-board trimmer, 4.7 R 2 W ballast
Hand-wired on one full-size ElectroCookie board. Documentation of the build, not a PCB.

Rob Lim

Sheet: /
File: fgl_rev_f.kicad_sch

Title: Fused Glass Lighting prototype (as built)

Size: A3	Date: 2026-10-15	**Rev: F**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1